

Wi-Fi Series

Data Sheet

USE_8188_FN

Wi-Fi Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	25/05/2026	MB	AD	Initial release

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Features

General

- Single-band Wi-Fi module supporting IEEE 802.11 b/g/n wireless connectivity
- Dimension: L × W × H: 12.2 mm × 13 mm × 1.5 mm (typical)
- Wi-Fi Interface: USB 2.0
- Supported Operating Systems: Android / Linux / Windows
- Operating Temperature: 0°C to 70°C
- Storage Temperature: -40°C to 85°C

MCU Platform

- Chipset: RTL8188FTV

WLAN Features

- Operates at ISM frequency band (2.4 GHz)
- Supports IEEE wireless standards:
IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, IEEE 802.11d, IEEE 802.11e, IEEE 802.11h,
IEEE 802.11i
- Enterprise level Wi-Fi security supporting WPA / WPA2 certification
- 1 Transmitter and 1 Receiver (1T1R) architecture supporting data rates up to 72.2 Mbps PHY rate
- Supports 802.11i security mechanisms including:
WPA, WPA2, WPA3, Open authentication, Shared key authentication, and Pairwise key authentication services

WLAN Interface

- USB interface for Wi-Fi connectivity

General Description:

The **USE_8188_FN** is a low-cost and low-power Wi-Fi module designed for embedded wireless networking applications. It is based on a highly integrated IEEE 802.11 b/g/n WLAN single-chip solution that combines MAC, baseband, and RF functionalities.

The module operates in the 2.4 GHz ISM band and provides a USB interface for easy integration with host systems. It integrates major Wi-Fi functional blocks such as USB interface, MAC, baseband (BB), RF front end, power amplifier, EEPROM, and power management circuits, reducing the number of external components required.

With its compact size and high integration, the **USE_8188_FN** module is suitable for IoT devices, portable electronics, and other embedded wireless applications.

For more information, reach out to us at info@unisemindia.com.

Pinout Diagram:

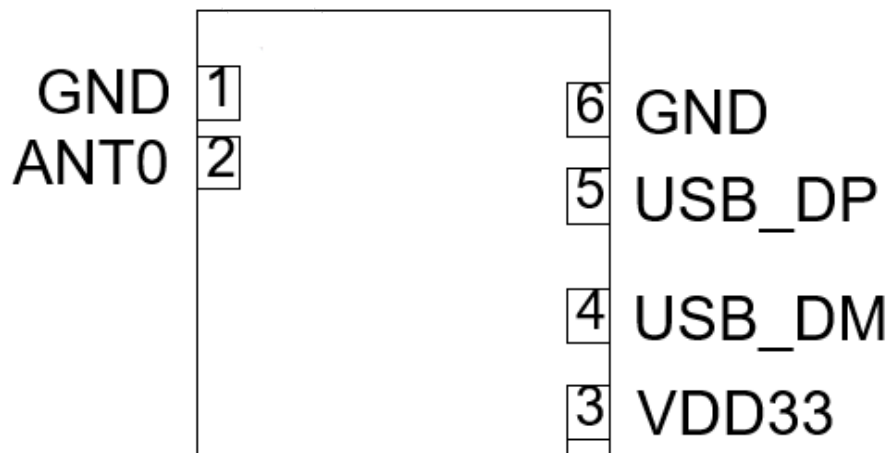


Fig1: Module Pinouts

Pin Descriptions:

NO.	Name	Type	Description
1	GND	—	Ground connections
2	ANT0	I/O	RF I/O port
3	VDD33	—	Main power voltage source input 3.3V
4	USB_DM	I/O	USB2.0 differential pair for WLAN
5	USB_DP	I/O	USB2.0 differential pair for WLAN
6	GND	—	Ground connections

The signal type codes are used in the table:

- I:INPUT
- O:OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	Min	Typ	Max	Unit
Supply Voltage	3.15	3.3	3.45	V
Operating Temperature	0	25	70	°C

Power Consumption

Power Consumption (Typical by using SWR)	TX Mode: (Throughput mode) 230 mA (MCS7/BW20)
	RX Mode: (Throughput mode) 130 mA (MCS7/BW20)

Thermal Characteristics

	MIN	Type	MAX	Unit
Junction Temperature	0		125	°C
θ_{JT} (°C/W)		0.43		°C/W
θ_{JA} (°C/W)		39.5		°C/W
θ_{JC} (°C/W)		21.2		°C/W

RF Characteristics :

Wi-Fi Radio Standard

Feature	Description
WLAN Standard	IEEE 802.11 b/g/n Wi-Fi compliant
Frequency Range	2.400 GHz ~ 2.4835 GHz (2.4 GHz ISM Band)
Number of Channels	2.4GHz: Ch1 ~ Ch14

Test Items	Typical Value	EVM
Output Power	802.11b / 11Mbps : 17 dBm $\pm$ 2 dB	EVM $\leq$ -10 dB
	802.11g / 54Mbps : 15 dBm $\pm$ 2 dB	EVM $\leq$ -25 dB
	802.11n / MCS7 : 14 dBm $\pm$ 2 dB	EVM $\leq$ -28 dB
Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	$\pm$ 20 ppm	
Test Items	TYP Test Value	Standard Value
Receive Sensitivity (11b, 20MHz) @8% PER	- 1 Mbps PER @ -91 dBm	$\leq$ -83
	- 2 Mbps PER @ -89 dBm	$\leq$ -80
	- 5.5 Mbps PER @ -87 dBm	$<$ -79
	- 11Mbps PER @ -85 dBm	$\leq$ -76
Receive Sensitivity (11g, 20MHz) @10% PER	- 6 Mbps PER @ -87 dBm	$\leq$ -85
	- 9 Mbps PER @ -86 dBm	$\leq$ -84
	- 12 Mbps PER @ -84dBm	$\leq$ -82
	- 18 Mbps PER @ -82dBm	$\leq$ -80
	- 24 Mbps PER @ -79 dBm	$\leq$ -77
	- 36 Mbps PER @ -75dBm	$\leq$ -73
	- 48 Mbps PER @ -71 dBm	$\leq$ -69
	- 54 Mbps PER @ -70 dBm	$\leq$ -68
Receive Sensitivity (11n, 20MHz) @10% PER	- MCS=0 PER @ -87 dBm	$\leq$ -85
	- MCS=1 PER @ -84 dBm	$\leq$ -82
	- MCS=2 PER @ -82 dBm	$\leq$ -80

	– MCS=3 PER @ -79 dBm	≤ -77
	– MCS=4 PER @ -75 dBm	≤ -73
	– MCS=5 PER @ -71 dBm	≤ -69
	– MCS=6 PER @ -70 dBm	≤ -68
	– MCS=7 PER @ -69 dBm	≤ -67
Maximum Input Level	802.11b : ≤ -8 dBm	
	802.11g/n : ≤ -20 dBm	

Module Diagram (Top View) :



Fig 2(a) : Module Diagram (Top view)

Module Diagram (Bottom View)

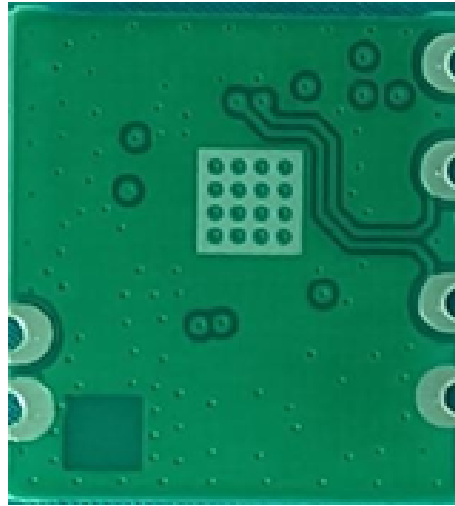


Fig 2(b) : Module Diagram (Bottom view)

Module Dimension Drawing:

- Module Dimension: $L \times W : 12.2 \text{ mm} \times 13 \text{ mm}$.

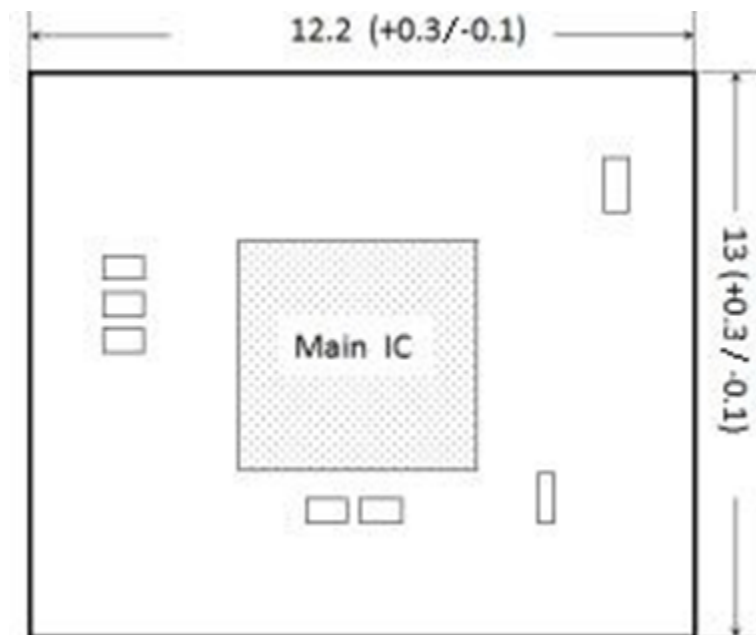


Fig 3(a) : Module Dimension $L \times W$

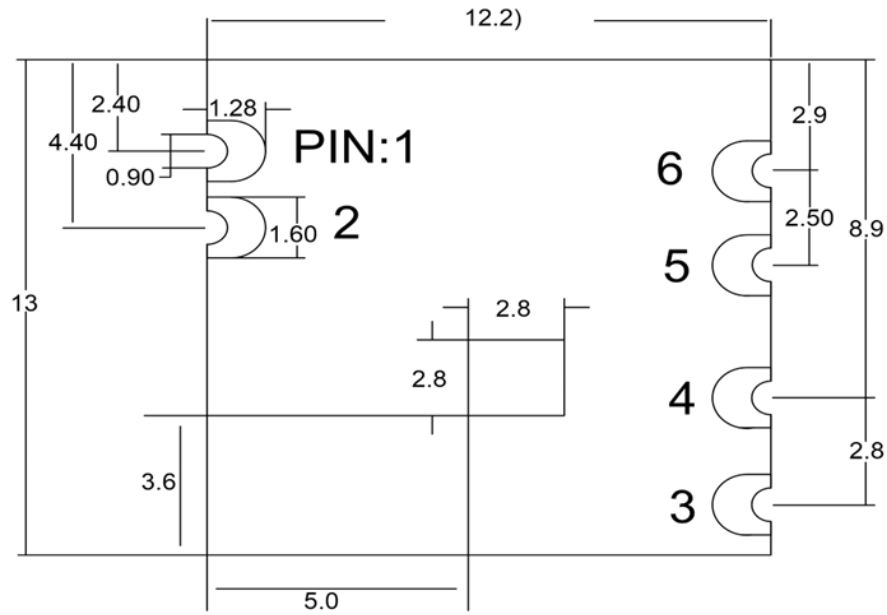


Fig 3(b) : Pad Dimensions

Product Dimension:

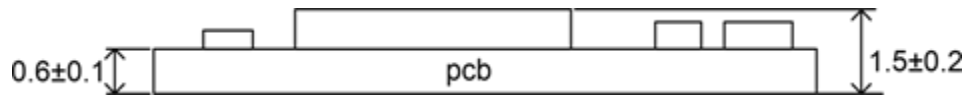


Fig 4 : Product Dimension

PCB thickness : 0.6 ± 0.1 mm

Product Height : 1.5 ± 0.2 mm

Recommended Footprint Dimensions :

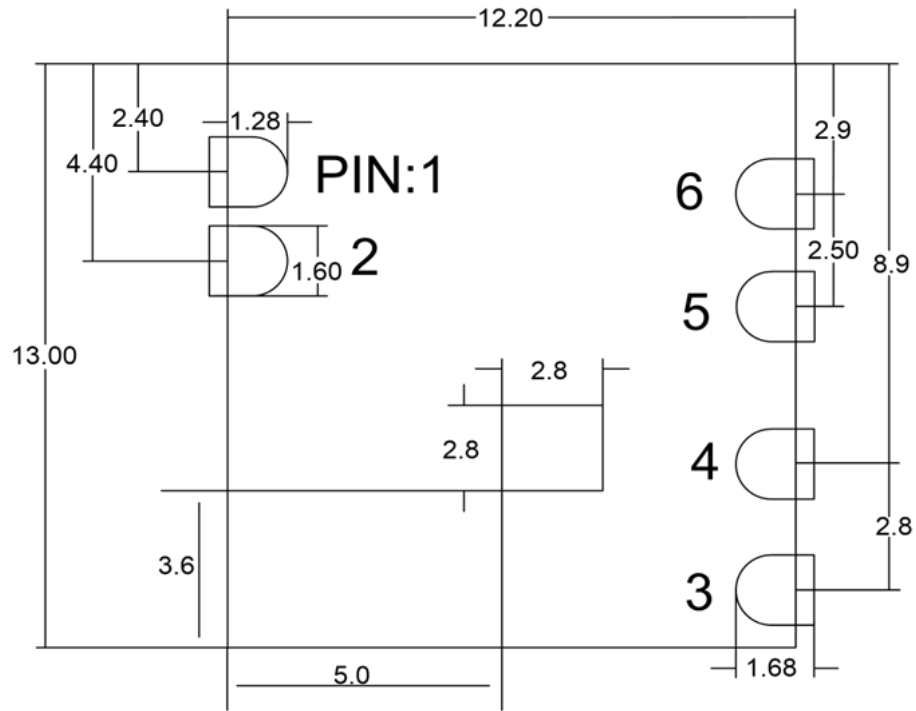


Fig 5 : Recommended Footprint Dimensions

Block Diagram :

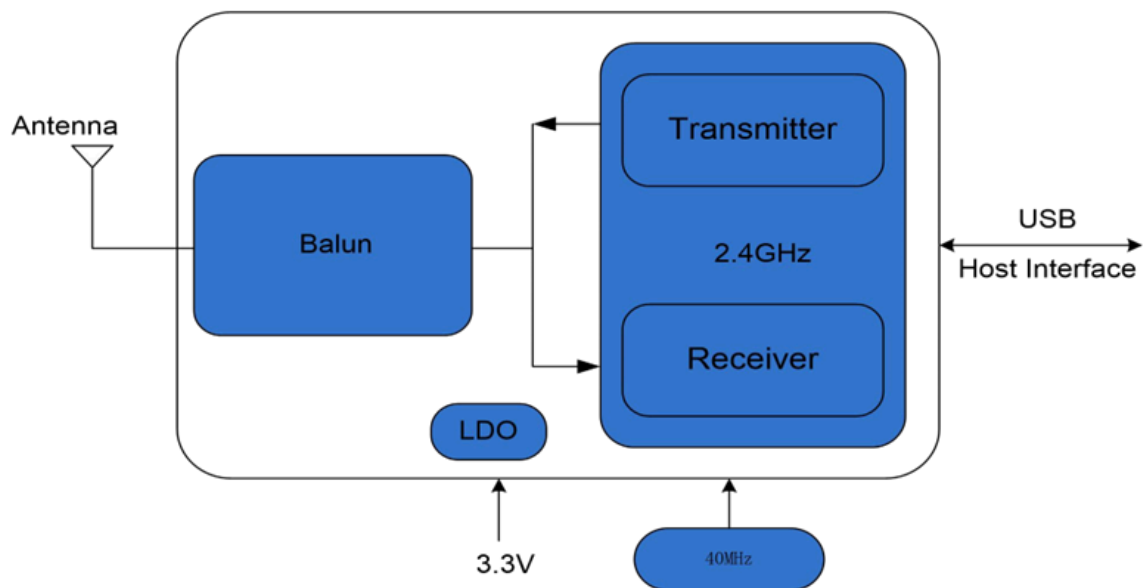


Fig 6 : Block Diagram

Reference Design :

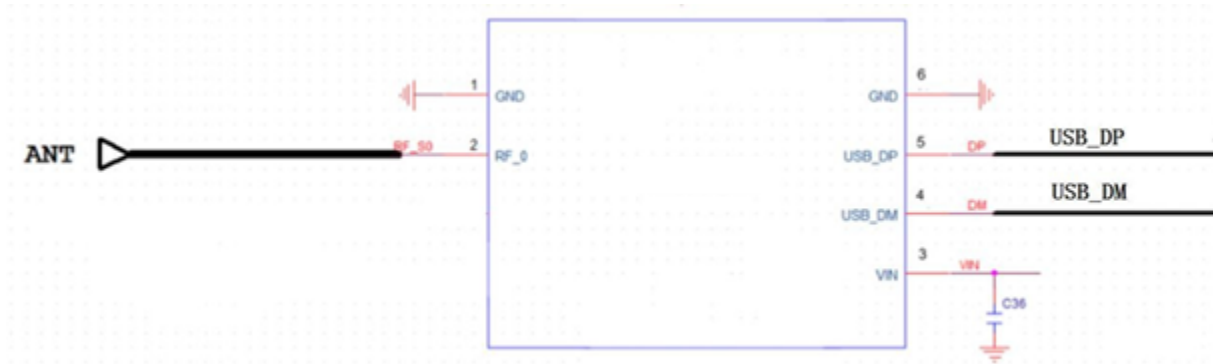


Fig 7 : Reference Design

Recommended Reflow Profile :

Referred to IPC/JEDEC standard.

Peak Temperature : <250°C

Number of Times : £2 times

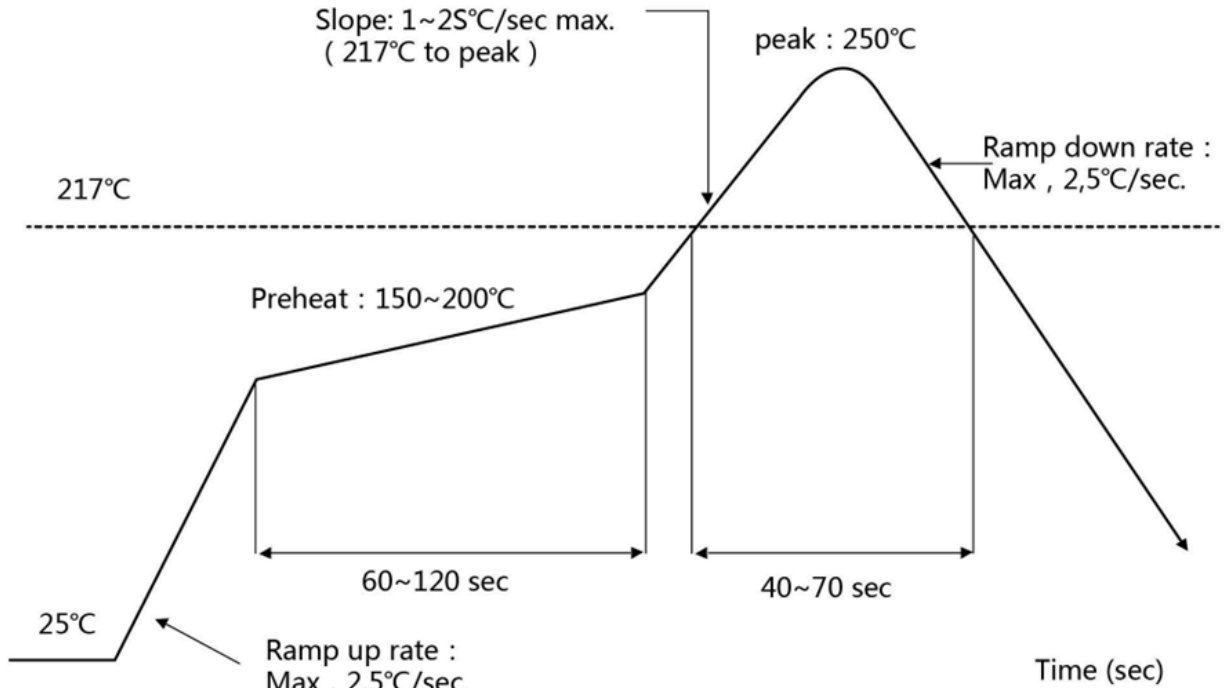
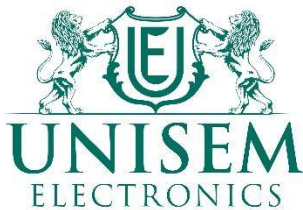


Fig 8 : Recommended Reflow Profile

For ordering USE_8188_FN please contact us on info@unisemindia.com.



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